

The Observer and the Vacuum: Zero Detection and the Ontological Loading of Absence

Juliet Zhong

Independent Researcher | London, United Kingdom | August 2026

ORCID: 0009-0006-5099-3671

Abstract

A region is called empty when nothing is detected in it. This paper examines what such a designation establishes and what it does not. Four companion papers established that the observer is the existential precondition of physical theory, that discrimination is realised electromagnetically, that observing systems sharing a radiative expression constitute a rendering domain individuated by mechanism rather than position, and that no domain-internal operation resolves a cosmological datum into an observer-independent remainder. Those results concerned what observation registers. This paper turns to what it fails to register, and takes the vacuum as the case where a null result has been carried furthest.

Two criteria share the name. The empirical vacuum is a region in which no particles are detected under a given mechanism, sensitivity and interval; the vacuum state $|0\rangle$ is the lowest-energy eigenstate of a specified Hamiltonian. They are not identical and serve different formal roles, and the continuity of the word between them permits an ontological intuition to travel from one context to the other unremarked. Neither criterion defines absence. The first is a domain-conditioned null result, from which ontological absence does not follow. The second presupposes the fields whose state it specifies, and is in addition not observer-independent: it varies with the observer's motion and admits no unique global form in a general spacetime. The concluding analysis traces how a term introduced to mark undetected content acquired the standing of a state, an energy, and a gravitational source. Falsification conditions are stated.

Keywords: vacuum; zero detection; ontological absence; observer dependence; rendering domain; vacuum state; Unruh effect; zero-point energy; cosmological constant; conceptual transfer

1. The Question Torricelli's Tube Left Open

1.1 A space above the mercury

In 1644, in Florence, Evangelista Torricelli sealed a glass tube at one end, filled it with mercury, inverted it into a basin, and watched the column fall until it stood at about seventy-six centimetres. Above the mercury there was a space. The tube had been full; now part of it was not. Nothing had been let in, and nothing had been taken out except the mercury that had descended.

What was in that space became one of the most consequential disputes of the century, and its structure is worth recovering, because the dispute was not about the experiment. Everyone agreed on what the apparatus did. The disagreement concerned what could be inferred from it. Torricelli's supporters held that the space contained nothing. Their opponents held that it contained a subtle matter too fine to be weighed, sensed, or otherwise registered by any means then available, and that the appearance of emptiness was a limitation of the instruments rather than a property of the space.

The opponents were, on the physics, wrong about the particular substance they proposed. They were not wrong about the logic of the objection. The apparatus could establish that no air was present, because the presence of air had detectable consequences that were absent. It could not establish that nothing was present, because no arrangement of that apparatus produced a signature of nothing. It produced only the absence of the signature of something. What the tube demonstrated was the removability of air. That the residue was empty was an inference, and the inference required a premise the experiment did not supply: that air was the only thing there was to remove.

Three and a half centuries later, the space above Torricelli's mercury is known to contain mercury vapour, and beyond that, the tube sits in a universe permeated by a microwave background, by neutrinos, and by whatever else has not yet been catalogued. The subtle-matter theorists lost their theory and won their objection. The objection has never been answered, because it is not the kind of objection that improved instruments answer. Every improvement in sensitivity lowers a threshold. No improvement in sensitivity converts a threshold into a floor.

This paper is about what happened to the word that Torricelli's contemporaries were arguing over, and about what physics has done with a designation whose original content was operational.

1.2 What the companion papers establish

Four companion papers precede this one. Their results are taken as established and are not re-derived; each is stated below in the form in which it will be used.

The first established that the observer is not a peripheral element of physical theory but its existential precondition (Zhong, 2026d). A reference frame cannot be physically instantiated without a system capable of discriminating between physical states and retaining a record. The Lorentz transformation,

deprived of such systems at both ends, is a map from the empty set to the empty set: formally valid, physically vacuous. And the Minkowski metric, whose invariance is conventionally taken as evidence of an observer-independent reality, presupposes the observer it is held to transcend. Physics has operated throughout with an existential presupposition it has not examined.

The second established the ground of the observer's discriminative capacity (Zhong, 2026e). Every known act of discrimination and registration, biological or instrumental or computational, is realised through electromagnetic interaction, and the convergence is not an accident of terrestrial technology: of the four interactions, only electromagnetism operates across the full range of scales from atomic to macroscopic, builds stable intermediate structures, supports cascade amplification, and yields recoverable records. What is registered is registered electromagnetically or it is not registered.

The third established that observing systems sharing a common physical expression of the radiative substrate constitute a rendering domain, within which mutual rendering is coherent and across which it is not (Zhong, 2026f). The criterion of domain membership is the mechanism through which discrimination is realised, not the position of the observer or the range of inputs its instruments register. Domains are therefore individuated by mechanism, and are not spatial regions. The boundary between them is not a wall that a sufficiently long journey or a sufficiently powerful telescope crosses.

The fourth carried these results into cosmology and asked what status attaches to data obtained by observers who cannot leave the domain in which they were generated (Zhong, 2026g). A cosmological datum has the form $D(O, I, G)$: a quantity conditioned on observer state, on instrument and procedure, and on the domain within which observation is possible at all. The question was whether any operation F available within the domain yields from D a unique remainder independent of O . The answer was negative, established by exhausting the three ways an exit might be attempted. Movement changes position and carries the domain along. Construction multiplies channels within the domain, since every instrument is built from its materials, calibrated against its standards, and read by its procedures. Inference establishes a decomposition only relative to a model built and validated within the domain, and no benchmark independent of the observer-domain coupling exists against which the residual could be checked. Two consequences were drawn. Reliability and observer-independence are distinct properties, and cosmology possesses the first in full measure without having established the second. And the Copernican principle is a condition of cosmological inference rather than one of its findings, since testing it would require comparing observations from positions outside the domain, and every available position is inside.

1.3 The question this paper asks

The fourth paper analysed what observation registers: temperatures, expansion rates, fluctuation amplitudes, all of them conditioned in the manner described. It did not ask about the other outcome. Every measurement has two possible results, and only one of them was examined.

When an instrument registers nothing, a claim is made about the world on the strength of it. The claim is not usually noticed as a claim, because a null result feels like the absence of a claim rather than the presence of one. But the inference from "this instrument registered nothing" to "there is nothing there" is a substantive step, and it is a step of exactly the kind the companion papers have been examining: it moves from a domain-conditioned outcome to a statement about what is the case independently of the domain.

The vacuum is the case in which this step has been taken furthest and with the least examination. What began as a designation for a region from which air had been removed became a region containing no matter, then a region containing no particles, then a physical state, then a state possessing energy, then a source term in the field equations of gravity, and finally a candidate explanation for the accelerating expansion of the universe. At no point in this progression was the original inference revisited. The term was carried forward, and each new theoretical role attached to it inherited the ontological standing the previous role had accumulated.

Two things must be said at the outset about what follows, because the argument is open to two misreadings and both are refused here.

The first is that this paper does not claim that physics is mistaken about the vacuum, that the measurements are unreliable, or that the theoretical treatment of vacuum states is defective. The treatment is extraordinarily successful, and nothing in what follows requires any calculation to be revised. What is examined is the relation between a term's operational content and the ontological load it has come to carry, which is a different question from whether the physics works.

The second is that this paper does not argue that the universe contains no vacuum, nor that vacuum is a fiction. Both would be stronger claims than the argument supports and neither is needed. The claim is narrower and, being narrower, is harder to defeat: an empirical vacuum designation is a domain-conditioned null result from which ontological absence does not follow, and the vacuum state of quantum field theory, which is not a null result but a mathematical definition, defines something other than absence. Neither route arrives at nothingness. What has been called vacuum, in every sense physics gives the word, is not nothing.

2. The Empirical Vacuum

2.1 What the designation records

Consider what is actually done when a region is certified as vacuum.

In a laboratory, gas is evacuated from a chamber and the residual pressure is measured, which is to say that a gauge responds to the flux of molecules striking a surface, and the response falls below some value. In astronomy, a line of sight is examined for absorption features, emission features, or scattering, and where these fall below the noise the column density is reported as an upper limit. In particle physics, a detector is operated over an interval and no events above threshold are recorded. In each case the

procedure has the same shape. An apparatus with a characterised response is exposed to a region for an interval, and its output does not exceed a stated value.

This is a measurement, and it is often an extraordinarily precise one. Modern ultra-high-vacuum systems reach pressures at which the residual number density is of order a hundred thousand molecules per cubic centimetre, which is a reduction of some fourteen orders of magnitude from atmospheric. Astronomical limits on the density of the intergalactic medium are similarly impressive as limits. Nothing in this analysis disputes the quality of the work.

What must be stated is the form of what has been established. The output of the procedure is: under this mechanism of interaction, at this sensitivity, over this interval, no signal above threshold. The designation "vacuum" is then applied to the region. The designation is a name for the outcome of the procedure, and the procedure has three parameters built into it. It has a mechanism, since the gauge responds to molecular impacts and the telescope responds to photons and neither responds to what does not couple to it. It has a sensitivity, since every threshold is finite. And it has an interval, since every observation is bounded in time.

An empirical vacuum, then, is a region in which no entity of a specified kind is detected, by a specified mechanism, above a specified threshold, within a specified interval. That is the complete content of the designation as the procedure delivers it.

2.2 Two propositions that are not the same

The step from that content to a claim about the world is a step from one proposition to another, and the two are not equivalent.

The first proposition is: no signal was registered. This is a fact about the outcome of a procedure, and it is as reliable as the procedure is.

The second proposition is: nothing is there. This is a claim about the region, and it does not follow from the first without an additional premise, namely that everything that could be there would have produced a signal.

That premise is never available in full. It could only be established by an enumeration of everything that might be present together with a demonstration that each member of the enumeration couples to the mechanism in use. The enumeration cannot be completed, because it would have to include what has not been catalogued, and the demonstration cannot be given for the uncatalogued portion, because there is nothing to demonstrate it about. What can be established, and routinely is, is the narrower and perfectly respectable claim that entities of specified kinds are absent above specified limits. Physics states this correctly in its technical literature, where upper limits are reported as upper limits. The slippage occurs in the transition to ontological language, where "no detected particles" becomes "empty space."

This is not a subtle point and it is not new. It is the elementary observation that a negative existential claim is not established by a failed search, and it holds for any negative existential claim whatever.

Stated at that level of generality it is a remark from the philosophy of science and not a result about physics, and if the argument stopped here it would deserve the response that everyone already knows this.

The argument does not stop here, and the next two moves are what give the observation purchase on the specific case.

2.3 The null result as a domain-conditioned datum

The first move is to locate the null result within the analysis the companion papers have already established.

A null result is a datum. It has the form $D(O, I, G)$ exactly as a positive measurement does: it is conditioned on the observer's state, on the instrument and its procedure, and on the domain within which observation occurs. The vacuum designation is the value of that datum when the value is zero, and being zero does not exempt it from the conditioning. If anything the conditioning is more consequential in the null case, because a positive result carries information about what produced it, some of which constrains the conditions of its own production, whereas a null result carries none.

The second companion paper's result now enters, and it sharpens the mechanism parameter into something specific. Discrimination in this domain is realised electromagnetically. Every instrument that has ever certified a region as empty did so through an electromagnetic mechanism: the pressure gauge through molecular impacts mediated by electromagnetic interaction, the telescope through photons, the particle detector through ionisation, scintillation, or charge collection, each of them electromagnetic at the point where the registration occurs. The convergence established in that paper was total across biological, instrumental and computational registration, and it holds here without exception.

So the content of a vacuum designation, stated with the mechanism made explicit, is: no electromagnetically coupled entity was registered here, above this threshold, within this interval. That is a considerably narrower claim than emptiness, and the narrowing is not a philosophical caveat added from outside. It is what the second companion paper established about the only mechanism by which any registration occurs.

The third move is the one that closes the escape route, and it is the fourth companion paper's theorem applied to the null case.

It might be held that although a single null measurement is conditioned, the conditioning can be removed by combining measurements: different instruments, different mechanisms in the loose sense of different detector technologies, different positions, different epochs, cross-validated against one another until the residual is the region itself. This is a reasonable proposal and it is what the practice does. The fourth companion paper answered it in general terms and the answer applies here without modification. Consistency among many determinations establishes that no single instrument or pipeline is responsible for the result. It does not establish that the class of determinations is free of a contribution common to all of them, because a common contribution is precisely what cross-validation among

members of the class cannot detect. Every member of the class registers electromagnetically. Every member is built, calibrated and read within the domain. The class has a common condition, and no operation performed among its members exhibits that condition as a separable term.

The conclusion for the null case follows in the same form as the general result. There is no operation F , available within the domain, that converts a domain-conditioned null result into an observer-independent statement of absence. Zero detections, however many instruments produce them and however far apart those instruments are placed, remain zero detections by the domain's mechanism. They do not accumulate into ontological absence, for the same reason that access does not accumulate into exit.

This is the paper's first result, and it is worth stating in the flattest available language. The reliability of a vacuum designation and the emptiness of the region designated are two different properties, and physics has established the first with great care while treating the second as though it followed.

3. The Distance Between the Vacuum and Nothingness

The result of the previous section is structural: it holds whatever the region actually contains. It is worth setting beside it what is known, on entirely conventional grounds, about what regions called empty do contain, because the gap between the designation and the ontology is not merely a logical possibility. It is a measured quantity, and it is large.

Three layers must be distinguished, and they differ in evidential standing. Conflating them would strengthen the appearance of the argument while weakening the argument itself.

3.1 What cannot be pumped out

The first layer is residual matter, and it is straightforwardly empirical.

No terrestrial procedure evacuates a region completely. Ultra-high-vacuum systems are limited by outgassing from the chamber walls, and as the pumping improves this source comes to dominate; the residual density falls but does not approach zero, and the practical limits are set by materials rather than by pumps. This is a technological fact and carries no ontological weight by itself.

The astronomical case is more instructive because the technological limits do not apply. The interstellar medium of the galaxy holds of order one atom per cubic centimetre. The intergalactic medium is far more dilute. Cosmic voids, the emptiest large structures known, have densities well below the cosmic mean, and in their interiors the mean atomic density falls to something of order one atom per cubic metre. At that density the mean separation between atoms is of order a metre, so that most individual cubic metres contain no atom at all: the figure is an average over a large volume and does not describe any particular one. Voids are genuinely and deeply empty of atoms, and there is no reason to soften this.

But atoms are not the only matter present. Cosmic rays traverse every region of the galaxy. The cosmic neutrino background, a relic of the first second, has a predicted number density of order a few hundred per cubic centimetre and is present everywhere; it has not been directly detected, and its existence rests on the same nucleosynthetic and cosmological reasoning that supports the rest of the standard cosmological model, but no volume is excluded from it. If the standard cosmological model is approximately correct, dark matter is distributed throughout the cosmic web at densities of order a fraction of a GeV per cubic centimetre in the solar neighbourhood, and is less concentrated than luminous matter, so that a region containing no atoms need not be free of it.

The last of these is the sharpest illustration of the section's point, and it deserves to be stated as such. Dark matter is inferred from gravitational effects and has never been registered by any electromagnetic instrument, which is what its name records: it does not couple to the mechanism by which every direct detection is performed. Whatever else it is, it is on this account a component of the universe that can be present in regions certified as empty and that the certifying instruments cannot see. Physics has therefore already encountered, within its own standard model of cosmology, an instance of the exact situation the previous section described in general terms: a region can be electromagnetically null and materially occupied, and the null designation records the mechanism rather than the region.

3.2 What cannot be shielded

The second layer is background radiation, and it is the best-measured item in the entire discussion.

The cosmic microwave background fills the universe. Its temperature is close to 2.725 K and its photon number density is of order four hundred per cubic centimetre, and it is not a signal arriving from a direction but a bath present at every point (Planck Collaboration, 2020a). No shielding excludes it, because any shield is at a temperature of its own and radiates accordingly; no region has been found without it, and the standard cosmological model gives no reason to expect one at the present epoch.

One qualification belongs here and it should be made in the paper rather than left for a referee to make. The microwave background is thermal radiation, and thermal radiation dilutes. The universe continues to expand, the background continues to cool, and its density continues to fall. So the statement that no region is free of it is a statement about the present epoch, and someone may reasonably respond that this reflects the current temperature rather than a structural impossibility. That response is correct as far as it goes, and it is the reason the argument of this paper does not rest on the microwave background. What the background establishes is an existential fact about the present universe: there is at present no region, anywhere, in which radiation is absent. It does not establish that such a region is impossible, and the argument does not need it to.

The contrast between the two layers is what earns the section its place. Matter is distributed in lumps with gaps between them, and regions containing no atoms are easy to find. Radiation is not distributed that way. It has no gaps. The observational asymmetry between the two is a fact of the present universe, whatever one concludes from it.

3.3 What cannot be removed

The third layer is theoretical, and its standing must be marked carefully, because it is where the largest claims in the literature are made and where the largest confusions occur.

In quantum field theory, the state of a field with no particles present is not a state in which the field is absent. The field is defined throughout the region; what is specified is which of its states obtains. In the lowest-energy state, the expectation value of the field vanishes while the expectation value of its square does not: $\langle 0 | \phi | 0 \rangle = 0$ and $\langle 0 | \phi^2 | 0 \rangle \neq 0$. The variance is non-zero. This is not a description of the field oscillating in time. It is a statement about the distribution of values that measurements of the field would yield, and about the fact that the distribution has width.

The origin of this width is the uncertainty relation between a field and its conjugate momentum, which forbids the simultaneous specification of both. It is therefore not a thermal effect and does not vanish with temperature, and it is not a dilution effect and does not vanish with expansion.

The empirical standing of this layer requires care. The variance itself is not directly observed; what are observed are effects that the standard treatment attributes to it. The Lamb shift in atomic hydrogen was measured in 1947 and is calculated with reference to the coupling of the electron to the quantised electromagnetic field (Lamb & Retherford, 1947). The Casimir force between conducting plates was predicted in 1948 and measured in the modern era to good precision (Casimir, 1948; Lamoreaux, 1997). The dynamical Casimir effect, in which a boundary condition modulated at relativistic speeds produces real photons, was observed in a superconducting circuit in 2011 (Wilson et al., 2011). These are among the best-tested predictions in physics.

They do not, however, license the inference that a definite quantity of energy resides in every cubic centimetre of space, and one prominent argument holds that they do not license even the weaker inference usually drawn. The Casimir force, on Jaffe's analysis, can be computed entirely in terms of the interaction between the charges and currents of the plates, with the coupling constant appearing explicitly and the zero-point energy of the free field never invoked; on that treatment the effect demonstrates the physics of quantum electrodynamics without demonstrating the reality of a vacuum energy density (Jaffe, 2005). The point is not that the standard account is wrong. It is that the effects establish that the theory works, and the further claim that the theory's vacuum is a reservoir of energy is a claim about interpretation which those effects do not by themselves settle. That distinction becomes the subject of Chapter 6.

What this layer does establish, and it is enough for the present purpose, is that the theory's own account of a region with no particles is not an account of a region with nothing in it. Whatever else the vacuum state is, it is a state of fields the theory has already specified. In the standard formulations of quantum field theory, the vacuum is not the absence of fields, and the theory contains no procedure that removes them.

3.4 What the three layers show together

Taken together the three layers give the measure of the gap. A region certified as empty contains, on entirely orthodox grounds, residual atoms in most cases and dark matter in all, a bath of relic photons and relic neutrinos everywhere, and fields in a state whose variance no operation reduces. The designation records that an electromagnetic instrument found nothing above threshold. The inventory is what is there.

None of this shows that a region of genuine absence is impossible. That is not the claim and it is not needed. What it shows is that in every case where the question has actually been examined, the gap between the designation and the contents has been large, and it has been closed in one direction only: each new investigation has added to the inventory. The subtle-matter theorists of the seventeenth century were wrong about mercury vapour and right about the structure of the inference, and the structure of the inference has not changed.

4. Two Criteria Under One Name

4.1 The two criteria

Physics uses the word "vacuum" under two distinct criteria, and the purpose of setting them side by side here is to analyse a definitional discontinuity. It is not to claim that the two are in contradiction, and it is not to claim that they have been reconciled. Neither claim would be correct.

The first criterion is the one inherited from classical physics and still in continuous use. A vacuum is a region containing no material medium. This is the sense carried by the constants of classical electrodynamics, where the speed of light in vacuum, the vacuum permittivity ϵ_0 and the vacuum permeability μ_0 characterise propagation in the absence of a material medium; it is the sense in which Maxwell's equations in vacuum are written; and it is the sense operative in every laboratory and every vacuum chamber. On this criterion, light passing through a region does not disqualify it. Radiation is what propagates through the vacuum; it is not what fills it.

The second criterion belongs to quantum field theory. The vacuum state $|0\rangle$ is the lowest-energy eigenstate of the Hamiltonian, equivalently the state annihilated by every annihilation operator, so that $a_k |0\rangle = 0$ for all modes k (Peskin & Schroeder, 1995; Weinberg, 1995). On this criterion the presence of a single photon disqualifies a state from being the vacuum, since a photon is an excitation of the electromagnetic field and a state containing one is not the lowest. The criterion is not about the absence of a medium but about the excitation level of the fields.

These two do not cut in the same place, and the divergence is easy to exhibit. Take a chamber evacuated of gas with sunlight passing through it. On the classical criterion this region is a vacuum, and it is precisely the configuration in which the vacuum speed of light is defined. On the quantum field theoretic criterion, the electromagnetic field in that region is manifestly not in its lowest state, so that

the same region satisfies the classical criterion while the field state within it is not the vacuum state. The two criteria take different logical subjects, a region in the first case and a state in the second, and the verdicts they return do not coincide.

4.2 Non-identical and differently employed

It would be a mistake to present this as an inconsistency in physics, and the mistake is worth naming so that the analysis is not misread as making it.

The two criteria answer different questions and serve different formal roles. The classical criterion answers a question about media: whether propagation through this region is modified by matter within it, which is what determines whether the free-space equations apply. The quantum criterion answers a question about states: which state of the field obtains, which is what determines what calculations of transition amplitudes take as their reference. Each is well defined within its own setting, each is used correctly by practitioners, and there is no situation in which a physicist is confused about which is meant. The retention of one word for two criteria has cost the practice nothing.

What it has permitted is something else, and it belongs to a different level of analysis. When a term migrates between theoretical settings while retaining its name, the ontological intuition attached to the name migrates with it, and does so without being restated or defended in the new setting, because nothing in the migration signals that a restatement is required. The reader of a text on classical electrodynamics acquires from "vacuum" the sense of a region with nothing in it. That sense is approximately harmless there, because on the classical criterion the claim being made is about the absence of a material medium and can be verified as such. When the same reader encounters the vacuum state of quantum field theory, the acquired sense travels intact, and it is now attached to an object it does not fit: a state of fields that are present, whose variance is non-zero, and whose energy is a contested quantity. Nothing in the notation warns of the transfer, because the notation is the same word.

This is the method of an earlier paper in this series applied to a second term. The examination of the boundary between matter and radiation found that two classification systems coexist in physics, the fermion/boson taxonomy of the Standard Model and the inherited matter/radiation vocabulary of the nineteenth century, each internally serviceable, and that their coexistence conceals an unreconciled classificatory inconsistency in which "radiation" spans both a fermion flux and a boson flux (Zhong, 2026a). The analysis there was not that physicists are confused, and it is not that here. It is that a term carrying an unstated ontological commitment can move between classification systems without the commitment being examined at the point of transfer, and that the examination is worth performing.

The point of performing it in the present case is that the term does not stop at two settings. It acquires a third role, and a fourth, and the load it carries at the end is not the load it was given at the start.

5. What $|0\rangle$ Defines

5.1 The definition escapes the argument of Chapter 2

The argument of Chapter 2 applies to the empirical criterion. It does not apply to the vacuum state, and this must be conceded without qualification before anything further is said, because an argument that appeared to defeat $|0\rangle$ by observational considerations would be defeating something other than $|0\rangle$.

The vacuum state is not defined by a failed detection. It is defined mathematically, as the lowest-energy eigenstate of a specified Hamiltonian, and its properties are derived rather than measured. No instrument certifies a state as $|0\rangle$, and no threshold enters the definition. Whatever is wrong with inferring absence from a null result, nothing of that kind is wrong here. The definition is exact, and it is one of the most productive definitions in the history of physics.

Two features of it, however, are consequential for the question this paper is asking, and both are internal to the definition rather than imposed on it from outside.

5.2 What the definition presupposes

The first is a matter of logical order.

To speak of the lowest-energy eigenstate, a Hamiltonian is required. To write a Hamiltonian, a field content is required: which fields exist, with what couplings, on what manifold. The vacuum state is therefore specified relative to a prior specification of the fields whose state it is. It is the answer to the question "which state of these fields is lowest?", and that question cannot be asked before the fields have been given.

The consequence is that quantum field theory has never defined a state of nothingness and has never claimed to. It defines the state of minimum excitation of a given field content. The fields are the input to the definition, not its output. This is not a limitation that the theory conceals; it is stated plainly in every textbook treatment, where the construction begins from a Lagrangian and proceeds to the vacuum, never the reverse. But the consequence is rarely drawn in the ontological register, and the consequence is this: the object that physics calls the vacuum is, by construction, a state of a specified field content. It is not ontologically prior to the fields it presupposes, since those fields had to be supplied before the state could be identified.

The word covers the excitation level and says nothing about the substrate. Whatever the fields are, and whatever their existence consists in, the vacuum state does not address it. The question of where the fields come from is not answered by quantum field theory because it is not asked; the theory takes them as given and computes. This is the boundary of the theory's competence, not a defect in it, and it is a boundary the theory itself draws.

5.3 The identification of a vacuum is not observer-independent

The second feature is more consequential still, and it is the point at which the results of the companion papers reappear from an unexpected direction. The identification of a state as the vacuum, although mathematical and untouched by every observational argument of Chapter 2, is nevertheless not observer-independent, and neither is the particle content with respect to which that identification is made. This is not a claim advanced by this paper. It is a standard result of quantum field theory, established in the 1970s and treated as elementary in the specialist literature ever since.

The result has two parts.

The first concerns accelerated observers in flat spacetime. Fulling showed that the canonical quantisation of a field in Minkowski space does not yield a unique particle notion: the quantisation carried out with respect to the time coordinate of a uniformly accelerated frame is unitarily inequivalent to the standard one (Fulling, 1973). Davies and Unruh drew out the physical content: a uniformly accelerated detector moving through the standard Minkowski vacuum responds as though immersed in a thermal bath, with temperature proportional to its proper acceleration (Davies, 1975; Unruh, 1976). The state is one state. The inertial observer characterises it as containing no particles; the accelerated observer's detector clicks. The disagreement is not a disagreement about the state, nor a matter of one observer being mistaken. Particle content, on this result, is not an attribute the state possesses independently of the observer with respect to whom it is defined.

It should be stated that the Unruh effect has not been directly observed. The temperature associated with attainable accelerations is minute, and the experimental proposals that exist are demanding. What the effect has is a secure theoretical standing within the standard formalism, derived rather than postulated, and a place in the textbook treatments of quantum field theory in curved spacetime (Birrell & Davies, 1982; Wald, 1994).

The second part concerns curved spacetime, and it is structural rather than dynamical. The privileged status of the Minkowski vacuum rests on the Poincaré symmetry of flat spacetime, which supplies a preferred timelike Killing vector and hence a preferred decomposition of field modes into positive and negative frequency. A general spacetime has no such symmetry and no such Killing vector. In its absence the decomposition is not unique, different choices yield unitarily inequivalent quantisations, and there is no preferred vacuum state to be found (Birrell & Davies, 1982; Wald, 1994). The vacuum of quantum field theory in curved spacetime is not a state the spacetime possesses. It is a state relative to a choice, and the spacetime of cosmology is not Minkowski space.

The bearing of this on the argument of the companion papers is direct, and it is worth stating carefully because it is the strongest single result the present paper has to offer.

The fourth companion paper established that a cosmological datum is conditioned on the observing domain and that no domain-internal operation resolves it into an observer-independent remainder. That argument proceeded from the conditions of observation. The result just described reaches a structurally

parallel conclusion from inside the formalism, without any premise about observers, rendering, or domains: what counts as the vacuum depends on the state of motion of the observer with respect to whom the field is quantised, and in a general spacetime the dependence cannot be removed by any choice that is not itself a choice. Two independent routes, one from the conditions of observation and one from the mathematics of quantisation, arrive at the same conclusion about the same object.

The convergence is not evidence for the companion papers' account, and it is not offered as such; a result of standard quantum field theory does not support an ontology that it neither states nor requires. What it does is remove a line of objection. The observer-dependence of the particle notion, and the non-uniqueness of vacuum selection in a general spacetime, are not exotic theses that this series of papers has introduced. It is a property of the vacuum that quantum field theory derived for itself, half a century ago, and has taught ever since.

5.4 What the two routes have in common

The empirical criterion and the mathematical criterion fail to deliver nothingness for different reasons, and the difference is worth marking because it prevents the two arguments from being run together.

The empirical criterion fails to deliver it because a null result is a domain-conditioned datum, and no operation within the domain converts it into a statement of absence. The failure is epistemic in form: it concerns what can be established.

The mathematical criterion fails to deliver it because the state it defines is the minimum-excitation state of a field content that had to be given in advance, and because the identification of that state is in addition relative to an observer's motion and, in general spacetimes, to a choice of quantisation. The failure is definitional in form: it concerns what has been defined.

What they have in common is the conclusion. Under neither criterion does physics possess a term for nothing. Under the first it possesses a term for what its instruments did not register. Under the second it possesses a term for the quietest available state of things that are there. The word "vacuum" has never named an absence in physics. It has named a threshold, and then a state.

6. From Absence of Detection to Physical Property: The Ontological Loading of Vacuum

The preceding chapters have established what the two criteria deliver. This chapter examines how a term whose original content was operational came to carry ontological weight, and what happened at each point where the weight increased.

The status of this chapter must be stated before it begins. It does not establish the claims of Chapters 2 and 5, which stand or fall on their own grounds. It does not offer the difficulties encountered at the end of the sequence as evidence that the sequence was mistaken. What it does is display the transfers themselves, so that the point at which a term acquired a commitment it had not previously carried can be located rather than assumed.

6.1 The first transfer: from designation to state

The starting point is the operational designation described in Chapter 2: a region in which the relevant entities are not detected. This is a classification applied to a region on the basis of a measurement outcome, and it carries no commitment about what the region contains beyond the negative result. Torricelli's space, the evacuated chamber and the interstellar sightline are all designations of this kind.

The first transfer occurs with the quantisation of fields. The vacuum ceases to be a classification applied to a region and becomes a state of a system: a vector in a Hilbert space, an eigenstate of an operator, an object with a definite mathematical characterisation. This is a substantial change in the type of thing being spoken of. A designation is a description of a region relative to a procedure; a state is an element of the theory's ontology, on the same footing as any other state the system can occupy.

The transfer was not marked, because the word did not change. And the consequence of leaving it unmarked is a persistent ambiguity between the two, visible in the way the results of Chapter 4 are discussed: "the vacuum" refers sometimes to a region and sometimes to a state, and the properties of the second are routinely attributed to the first. A region of intergalactic space is described as having a vacuum energy density; what has an energy is a state, and whether that state obtains in that region is a further question which the description passes over.

6.2 The second transfer: from state to energy

The second transfer assigns the state an energy.

The formal basis is direct. Each mode of a quantised field contributes $\frac{1}{2}\hbar\omega$ to the energy of the lowest state, and the sum over modes is the zero-point energy. That the sum diverges was recognised immediately, and the standard response is that in the absence of gravity only energy differences are physically significant, so that the divergent constant may be removed by normal ordering or absorbed by renormalisation without observable consequence.

This response is entirely successful within its scope, and the scope is worth naming precisely: the standard non-gravitational observables invoked in this context do not require an independently measurable absolute vacuum-energy offset. Each of them is a difference. The Lamb shift is a difference between energy levels. The Casimir force is a difference between the configuration with plates and the configuration without them. The dynamical Casimir photons are produced by a time-dependent boundary condition, and what is calculated is a rate. The theory does not require the absolute value to be defined, and it takes no position on whether the divergent sum corresponds to anything.

The ontological transfer occurs when the removed constant is nevertheless spoken of as a quantity that is there. This move is not part of the calculation and is not required by it. It is an interpretation of a term that the calculation discards, and it has been contested within physics: on Jaffe's analysis the Casimir force can be derived without invoking the zero-point energy of the free field at all, which does not show the standard account to be wrong but does show that the effect fails to establish the reality of

the discarded term (Jaffe, 2005). The philosophical literature has made the same point in more general form, arguing that the vacuum energy density of quantum field theory is not straightforwardly a physical quantity of the kind that the cosmological discussion assumes (Rugh & Zinkernagel, 2002).

At this stage of the sequence, then, the term carries: a state, and an energy attributed to the state whose status as a physical quantity is not settled by the calculations that succeed.

6.3 The third transfer: from energy to gravitational source

The third transfer is the one that makes the difference visible, because gravity is the one place where the absolute value cannot be discarded.

In general relativity every form of energy gravitates. If the vacuum possesses an energy density then it enters the stress-energy tensor, and a Lorentz-invariant vacuum energy has the equation of state of a cosmological constant: an energy density that does not dilute as space expands. Zel'dovich set out the identification, and the difficulty it produces has been the subject of a substantial literature since (Zel'dovich, 1968; Weinberg, 1989).

The difficulty is that the magnitude estimated from the field-theoretic side and the magnitude inferred from cosmological observation do not agree, and the discrepancy is not marginal. It should be described carefully. The theoretical estimate is not a single well-defined number: it depends on the cut-off scale adopted, on the regularisation scheme, and on which fields are included, and quoting a specific number of orders of magnitude gives the estimate a definiteness it does not have. What can be said without depending on any of these choices is that natural estimates from the field-theoretic side exceed the observationally inferred value of the cosmological constant by an enormous margin under any of the standard treatments, and that no accepted mechanism accounts for the difference (Weinberg, 1989; Rugh & Zinkernagel, 2002). The observational side is by contrast well determined: the accelerating expansion was established from supernova distances (Riess et al., 1998; Perlmutter et al., 1999) and the cosmological parameters have been measured to high precision (Planck Collaboration, 2020b).

The point of examining this here is not that the discrepancy proves anything about the vacuum's ontological status. It does not, and a substantial fraction of theoretical physics is directed at resolving it by means that leave the standard account intact. The point is that it is located exactly where the sequence predicts a problem would appear. The first two transfers took place in settings where the absolute value of the vacuum energy played no role and where nothing tested whether the term denoted a quantity. The third transfer placed the term in the one setting where the absolute value cannot be avoided, and the theory's largest unresolved quantitative discrepancy appeared at that step. That is not a proof. It is a diagnosis of where the load was placed.

6.4 The accumulated load

The sequence can now be stated in one line, and the statement is the substance of this chapter.

A term introduced to record that a procedure detected nothing came to name a state; the state was assigned an energy that no non-gravitational measurement requires to be defined; and the energy was made a source term in the field equations of gravity and a candidate explanation for the large-scale behaviour of the universe.

Each step was individually reasonable, and each was taken by physicists doing careful work for good reasons. What was never done was to return to the beginning and ask what the term had been introduced to record. The designation recorded a threshold. Four steps later, the same word denoted a quantity of energy in every cubic centimetre of space, driving cosmic acceleration.

This is what the ontological loading of a term consists in, and it is why the analysis of Chapter 4 was not a pedantic point about vocabulary. A word that migrates between theoretical settings carries its accumulated commitments forward and acquires new ones at each stage, and the commitments are never audited, because auditing them would require noticing that a migration had occurred, and nothing in the notation records it.

7. Consequences and Falsification

7.1 What follows

Two consequences require statement, and both are narrower than the discussion that produced them.

The first concerns how null results should be classified. In current practice a null result is reported as an upper limit in the technical literature and translated into an emptiness claim in the interpretive one, and the translation is invisible because the same word does both jobs. On the analysis given here, a null result is a datum of the form $D(O, I, G)$ whose value is zero, and it is conditioned on the observing domain exactly as a positive result is. What it establishes is that entities of a specified kind, coupling to a specified mechanism, are absent above a specified threshold. What it does not establish is absence. The distinction has practical content: it identifies which questions further sensitivity will settle, namely questions about entities that couple to the mechanism in use, and which it will not, namely questions about what is there.

The second concerns the standing of the vacuum in cosmological reasoning. The vacuum energy is treated in the cosmological literature as a physical quantity whose value is to be determined, and the difficulty is understood as a discrepancy between two determinations of it. On the analysis given here, the prior question is whether the term denotes a quantity of that kind, and the answer is not established by the successes that made the term indispensable, since those successes are all differences and none of them tests the absolute value. This does not resolve the difficulty and is not offered as resolving it. It relocates it: the question is not only why the numbers disagree but whether the object whose numbers disagree has been shown to be an object.

Beyond these, the general result of the paper connects to the series that precedes it. The four companion papers established that physical theory presupposes an observer, that discrimination is

realised electromagnetically, that domains are individuated by mechanism, and that a datum is not resolvable within a domain into an observer-independent remainder. This paper adds the case of the null datum, and finds that it behaves as the general result predicts: zero detections do not accumulate into absence, for the same reason that access does not accumulate into exit. And it finds, in the observer-dependence of the vacuum state, that quantum field theory reached a structurally parallel conclusion about the same object by a route of its own.

7.2 Falsification

Each condition below is stated so that its satisfaction defeats a result of this paper rather than requiring its adjustment.

First, an operational criterion for ontological absence that does not consist in a null detection. The argument of Chapter 2 holds that the designation of a region as empty is the outcome of a procedure with a mechanism, a threshold and an interval, and that no such procedure establishes absence. A criterion by which a region could be certified as containing nothing, and which did not reduce to the failure of some instrument to register something, would defeat that argument directly. Criteria which reduce to null detection by a broader class of instruments do not satisfy this condition, and the condition is stated so as to exclude them.

Second, a demonstration that combination of null results across mechanisms yields an unconditioned absence claim. The argument holds that consistency among null determinations establishes that no single instrument is responsible, and not that the class is free of a common condition. An explicit separation of the common condition, validated against a benchmark not itself constructed within the domain, would end the result.

Third, a formulation of quantum field theory in which the vacuum state is defined without prior specification of a field content. The argument of Chapter 5.2 holds that the vacuum is the minimum-excitation state of fields given in advance, and that the theory therefore defines no state of nothingness. A construction that derived the field content rather than assuming it, and identified a vacuum with respect to no antecedent specification, would defeat that argument.

Fourth, a demonstration that the vacuum state is observer-independent after all. The argument of Chapter 5.3 rests on the non-uniqueness of the particle notion under inequivalent quantisations and on the absence of a preferred timelike Killing vector in a general spacetime. A demonstration that a preferred vacuum can be singled out in a general spacetime on physical grounds, without an additional choice, would remove the second route to the paper's conclusion, leaving the first standing but unsupported by it.

Fifth, a measurement demonstrating sensitivity to an absolute vacuum-energy offset by non-gravitational means. The analysis of Chapter 6.2 holds that the non-gravitational observables adduced in support of vacuum energy are differences, and that none of them requires an absolute offset to be defined. A measurement sensitive to the absolute value, independent of gravitational effects, would

show the term to denote a quantity in the way the third transfer assumed, and would defeat the diagnosis of Chapter 6.

7.3 The position

What remains is to state the position without qualification.

A region is called empty when an instrument built within one rendering domain, responding through the mechanism by which that domain discriminates, registers nothing above a threshold within an interval. That outcome is reliable and it is conditioned, and no operation available within the domain converts it into a statement about what the region contains.

Where physics has replaced the designation with a definition, the definition is not a definition of absence. The vacuum state is the minimum-excitation state of a field content specified in advance; it presupposes what it is a state of; and it is not the same state for observers in different states of motion, nor uniquely determined in a spacetime without the symmetry that flat space supplies.

Between these two, the word acquired an energy and then a gravitational role, and the largest unresolved quantitative discrepancy in the subject appeared at the step where the absolute value could no longer be discarded.

Physics has never had a term for nothing. It has had a term for what was not registered, and a term for the quietest state of what is there, and it has used one word for both. Torricelli's opponents were wrong about the substance and right about the inference, and the space above the mercury was never empty.

Acknowledgements

The author acknowledges the use of AI tools (Claude and ChatGPT) in the preparation of this manuscript. These tools were employed as supportive instruments for language refinement, structural organisation, and clarity improvement of the technical exposition. All scientific ideas, modelling choices, and interpretations presented in this work are the sole responsibility of the author. The use of AI did not involve any generation of experimental data or alteration of underlying physical assumptions, and all content was reviewed and validated by the author prior to submission.

Declarations

Funding: This research received no external funding.

Conflicts of interest: The author declares no conflicts of interest.

Data availability: No new observational data were generated or analysed in this study. All referenced datasets are publicly available from the sources cited.

Author contributions: Juliet Zhong: conceptualisation, formal analysis, writing.

References

- Birrell, N. D., & Davies, P. C. W. (1982). *Quantum Fields in Curved Space*. Cambridge University Press.
- Casimir, H. B. G. (1948). On the attraction between two perfectly conducting plates. *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, 51, 793–795.
- Davies, P. C. W. (1975). Scalar production in Schwarzschild and Rindler metrics. *Journal of Physics A: Mathematical and General*, 8, 609–616.
- Fulling, S. A. (1973). Nonuniqueness of canonical field quantization in Riemannian space-time. *Physical Review D*, 7, 2850–2862.
- Jaffe, R. L. (2005). Casimir effect and the quantum vacuum. *Physical Review D*, 72, 021301.
- Lamb, W. E., & Retherford, R. C. (1947). Fine structure of the hydrogen atom by a microwave method. *Physical Review*, 72, 241–243.
- Lamoreaux, S. K. (1997). Demonstration of the Casimir force in the 0.6 to 6 μm range. *Physical Review Letters*, 78, 5–8.
- Perlmutter, S., et al. (1999). Measurements of Ω and Λ from 42 high-redshift supernovae. *The Astrophysical Journal*, 517, 565–586.
- Peskin, M. E., & Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley.
- Planck Collaboration (2020a). Planck 2018 results I: Overview and the cosmological legacy of Planck. *Astronomy & Astrophysics*, 641, A1.
- Planck Collaboration (2020b). Planck 2018 results VI: Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.
- Riess, A. G., et al. (1998). Observational evidence from supernovae for an accelerating universe and a cosmological constant. *The Astronomical Journal*, 116, 1009–1038.
- Rugh, S. E., & Zinkernagel, H. (2002). The quantum vacuum and the cosmological constant problem. *Studies in History and Philosophy of Modern Physics*, 33, 663–705.
- Unruh, W. G. (1976). Notes on black-hole evaporation. *Physical Review D*, 14, 870–892.
- Wald, R. M. (1994). *Quantum Field Theory in Curved Spacetime and Black Hole Thermodynamics*. University of Chicago Press.
- Weinberg, S. (1989). The cosmological constant problem. *Reviews of Modern Physics*, 61, 1–23.

Weinberg, S. (1995). *The Quantum Theory of Fields, Volume 1: Foundations*. Cambridge University Press.

Wilson, C. M., Johansson, G., Pourkabirian, A., Simoen, M., Johansson, J. R., Duty, T., Nori, F., & Delsing, P. (2011). Observation of the dynamical Casimir effect in a superconducting circuit. *Nature*, 479, 376–379.

Zel'dovich, Y. B. (1968). The cosmological constant and the theory of elementary particles. *Soviet Physics Uspekhi*, 11, 381–393.

Internal References:

Zhong, J. (2026a). The Boundary Between Matter and Radiation: An Unresolved Classificatory Inconsistency in Modern Physics. SSRN. <https://doi.org/10.2139/ssrn.7186058>

Zhong, J. (2026b). Strings as the Ontology of Radiation: Dissolving String Theory's Fifty-year Impasse. SSRN. <https://doi.org/10.2139/ssrn.7260061>

Zhong, J. (2026c). The Ontological Boundary of Quantum Probability: Pre-Collapse Possibility versus Realised Physical Fact. SSRN. <https://doi.org/10.2139/ssrn.7300039>

Zhong, J. (2026d). [P1] The Observer and Spacetime: On the Existential Precondition of Physical Theory. SSRN. <https://doi.org/10.2139/ssrn.7316678>

Zhong, J. (2026e). [P2] The Observer and Consciousness: On the Ontological Ground of Measurement. Manuscript.

Zhong, J. (2026f). [P3] Rendering Isolation: On the Structural Incommensurability of Observing Domains. Manuscript.

Zhong, J. (2026g). [P4] The Observer and the Cosmological Datum: On the Ontological Invariance of the Observing Domain. Manuscript.